understand with a little effort, but just difficult enough that one is pleased with oneself for understanding it, and anxious to pass it on to others. It is a good 'meme'. If you learn about Hamilton not from reading him, but from a conversation in a pub, the chances are very high that you'll hear about nothing except haplodiploidy. Nowadays every textbook of biology, no matter how briefly it covers kin selection, is almost bound to devote a paragraph to 'relatedness'. A colleague, who is now regarded as one of the world's experts on the social behaviour of large mammals, has confessed to me that for years he thought that Hamilton's theory of kin selection *was* the relatedness hypothesis and nothing more! The upshot of all this is that if some new facts lead us to doubt the importance of the relatedness hypothesis, people are apt to think that this is evidence against the whole theory of kin selection. It is as if a great composer were to write a long and profoundly original symphony, in which one particular tune, briefly tossed out in the middle, is so immediately catchy that every barrow-boy whistles it down the streets. The symphony becomes identified with this one tune. If people then become disenchanted with the tune, they think that they dislike the whole symphony.

Take, for example, an otherwise useful article by Linda Gamlin on naked mole rats recently published in the magazine *New Scientist*. It is seriously marred by the innuendo that naked mole rats and termites are in some way embarrassing for Hamilton's hypothesis, simply because they are not haplodiploid! It is hard to believe that the author could possibly have even seen Hamilton's classic pair of papers, since haplodiploidy occupies a mere four of the fifty pages. She must have relied on secondary sources—I hope not *The Selfish Gene*.

Another revealing example concerns the soldier aphids that I described in the notes to Chapter 6. As explained there, since aphids form clones of identical twins, altruistic self-sacrifice is very much to be expected among them. Hamilton noted this in 1964 and went to some trouble to explain away the awkward fact that—as far as was then known—clonal animals did not show any special tendency towards altruistic behaviour. The discovery of soldier aphids, when it came, could hardly have been more perfectly in tune with Hamilton's theory. Yet the original paper announcing that discovery treats soldier aphids as though they constituted a difficulty for Hamilton's theory, aphids not being haplodiploid! A nice irony.

When we turn to termites—also frequently regarded as an embarrassment for the Hamilton theory—the irony continues, for Hamilton himself, in 1972, was responsible for suggesting one of the most ingenious theories about why they became social, and it can be regarded as a clever analogy to the haplodiploidy hypothesis. This theory, the cyclic inbreeding theory, is commonly attributed to S. Bartz, who developed it seven years after Hamilton originally published it. Characteristically, Hamilton himself forgot that he had thought of the 'Bartz theory' first, and I had to thrust his own paper under his nose before he would believe it! Priority matters aside, the theory itself is so interesting that I am sorry I did not discuss it in the first edition. I shall correct the omission now.

I said that the theory was a clever analogue of the haplodiploidy hypothesis. What I meant was this. The essential feature of haplodiploid animals, from the point of view of social evolution, is that an individual can be genetically closer to her sibling than to her offspring. This predisposes her to stay behind in the parental nest and rear siblings rather than leaving the nest to bear and rear her own offspring. Hamilton thought of a reason why, in termites too, siblings might be genetically closer to each other than parents are to offspring. Inbreeding provides the clue. When animals mate with their siblings, the offspring that they produce become more genetically uniform. White rats, within any one laboratory strain, are genetically almost equivalent to identical twins. This is because they are born of a long line of brother–sister matings. Their genomes become highly homozygous, to use the technical term: at almost every one of their genetic loci the two genes are identical, and also identical to the genes at the same locus in all the other individuals in the strain. We don't often see long lines of incestuous matings in nature, but there is one significant exception—the termites!